

CHAITANYA MAHAJAN

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EDUCATION

University of Colorado - Boulder
Master's, Computer Science

August 2024 - May 2026
GPA: 4.0

Indian Institute of Technology Mandi
Bachelor's, Computer Science

July 2019 - May 2023
GPA: 7.78

PROFESSIONAL EXPERIENCE

CableLabs

Computer Science Technologist Intern

Louisville, CO

May 2025 - August 2025

- Built a scalable pipeline to collect Upstream and Downstream PNM (Proactive Network Monitoring) data from Modems and CMTS using NETSNMP and Python.
- Integrated the pipeline with Victoria Metrics and Grafana for long term storage and visualization respectively, deployed the pipeline on a Kubernetes cluster.
- Used Docker for containerization; GitLab for version control, to manage the CI/CD pipeline
- Co-authored a research paper titled 'A Lightweight Attention-Guided Convolutional Neural Network for Explainable Multiclass Fault Classification in 1 Dimensional Signal Data'.

Explorin Academy

DSA Instructor

Roorkee, UT, India

February 2024 - April 2024

- Taught Data Structure and Algorithms, Object Oriented Programming Principles (OOPS) to Undergraduate students at the Roorkee Institute of Technology using C++
- Conducted Introductory Programming sessions for an audience of 450+ students at the College of Engineering Roorkee.

Data Sailors

Machine Learning Engineer

Jammu and Kashmir, India

November 2023 - January 2024

- Built and optimized ETL pipeline for collection and forecasting of energy consumption of commercial buildings
- Used hybrid LSTM models to achieve 95 % accuracy on prediction of IIT Jammu's Academic block energy consumption.
- Designed and implemented interactive Power BI dashboards to visualize forecasted energy trends against actual consumption, enabling stakeholders to identify efficiency gaps and make data-driven operational decisions.

Edfora Infotech Pvt. Ltd.

Front End Web Developer

New Delhi, DL, India

July 2023 - October 2023

- Integrated a scalable comment service functionality into Glorifire and Watson using Angular and TypeScript, facilitating direct communication between students and the Edfora Team.
- Styled responsive UI components with HTML/CSS, resulting in increased user feedback and actionable insights for platform improvement.
- Spearheaded the front-end development for myBatchPlan Stakeholder by translating high-fidelity Figma mockups into pixel-perfect code using Angular and Bootstrap. Built a user-friendly "Todo List" feature and management dashboard with strict TypeScript safety, directly streamlining batch management and boosting teacher engagement.

Indian Institute of Technology, Mandi

Teaching Assistant — Data Science

Mandi, HP, India

August 2022 - November 2022

- Conducted weekly Doubt Sessions, Vivas and graded Assignments
- Mentored 6 groups of students for their Course Project.

Siemens

Deep Learning Research Intern | Computer Vision

Remote

January 2022 - June 2022

- Optimized the AI4SAFETY team's video analysis pipeline and integrated it with Activity Recognition and Sequence Anomaly Detection models
- Conducted a comprehensive literature survey on various Machine Learning techniques, and implemented the Dynamic Inference Network model, achieving 94 % accuracy in Activity Recognition tasks
- Implemented Auto Encoders and achieved a 92 % accuracy in Sequence Anomaly Detection tasks
- Developed a video output generation pipeline that seamlessly annotated actors in the video and their corresponding activities, reducing post-production time and enhancing overall video quality.

PROJECTS & OUTSIDE EXPERIENCE

CISCO - LLM Triage Assistant

Masters Capstone Project | RAG System | Python, LangChain, Hugging Face, AWS (S3), ChromaDB

August 2025 - May 2026

- Architected a Retrieval-Augmented Generation (RAG) system using LangChain to orchestrate document retrieval and answer generation, automating root cause analysis for network infrastructure logs.
- Leveraged Hugging Face libraries and Google Vertex AI Workbench to fine-tune Open Source LLM Llama-3.2 models on GCP, leveraging high-performance GPUs to optimize the model for interpreting proprietary Cisco log formats (Syslog, gRPC).
- Developed a "Context Selection" algorithm to optimize LLM context windows, filtering 4,000+ line log files down to <150 high-signal lines by clustering error bursts and tail events.
- Engineered a custom ETL pipeline that normalized unstructured logs (gRPC, Syslog) and stored processed log data in AWS S3 buckets for scalable retrieval and archival.
- Implemented a Vector Search engine (ChromaDB) using sentence transformers to cluster historical failures, enabling automated deduplication of known bugs and novelty detection for new incidents.

LLM-Based Movie Recommendation System

Python, OpenAI API, GCP

- Utilized the OpenAI API to fine-tune a GPT-4o model on gold-standard recommendation datasets, tailoring the model's tone and accuracy for specialized content suggestions.
- Deployed data preprocessing pipelines on Google Cloud Platform (GCP) to clean and structure the Netflix Prize dataset, ensuring high-quality input features for the fine-tuning process.
- Conducted rigorous testing comparing baseline zero-shot models against the fine-tuned version, achieving over 10x improvement in key metrics such as Precision@3, Recall@3, and F1@3.

Clinic Appointment & Notification System | Next.js (React), FastAPI, PostgreSQL, Resend API

<https://dr-archana-gupta-clinic.vercel.app/>

- Engineered a responsive, state-driven patient portal for my mother's eye clinic, using React and Next.js, implementing React Hooks (useState, useEffect) to manage real-time booking flows and administrative dashboard views.
- Architected a cloud-native backend on Render utilizing FastAPI Middleware to implement secure CORS policies, enabling authenticated cross-origin communication with a decoupled Vercel frontend.
- Optimized database performance and cross-cloud compatibility by implementing a Supabase Transaction Pooler, resolving IPv4/IPv6 networking constraints and reducing connection overhead for PostgreSQL persistence.
- Developed an automated notification pipeline using the Resend API to deliver instant booking confirmations and administrative alerts, ensuring 100% delivery of time-sensitive patient data.
- Implemented an operational "warm-start" strategy using a scheduled Cron Job heartbeat to mitigate PaaS sleep cycles, maintaining high availability and reducing initial request latency by over 90%.

TECHNICAL SKILLS

Technical Skills: Python, C/C++, Java, HTML/CSS, JavaScript, TypeScript, SQL, PHP

Developer Tools: Jupyter Notebook, VS Code, Bitbucket, Gitlab, JIRA, Agile, Linux/Unix, GitHub

Technologies/Frameworks: AngularJS, React, Next.js, FastAPI, Node.js, MySQL, PostgreSQL, Victoria Metrics, Hadoop, Snowflake, Postman, MongoDB, Apache Kafka, Pytorch, Tensorflow, OpenCV, LangChain, CUDA, Apache Spark, Grafana, Prometheus, NETSNMP, Jenkins, Kubernetes, Docker, Helm, GCP